Name:_____ Period:_____

The Atom That Would Not Die: A Carbon Cycle Mystery

Learning Intention: Today we will be learning how carbon moves between the atmosphere, biosphere, hydrosphere, and geosphere so we can explain why the amount of carbon dioxide in our air is changing.

Success Criteria:

- I can identify the process that moves carbon from one reservoir to another.
- I can use CO2 data to explain why the amount of carbon dioxide in the atmosphere is increasing.

A carbon atom named **Count Carbonula** has been on the move for 300 million years. It has been inside a volcano, a giant swamp plant, a lump of coal, and a jack-o'-lantern. Your job is to follow **Count Carbonula** through each case file, figure out how it moved, and crack the secret code.

Part 1: Carbon Case File

Read this page carefully. You will need it for every question in the packet.

Carbon atoms are never created or destroyed in the carbon cycle. They move from one place to another, over and over. The places where carbon is stored are called **reservoirs**.

Reservoir	What it is	How carbon is stored there
Atmosphere	The air around Earth	Carbon dioxide (CO_2) gas
Biosphere	All living and once-living things	Sugar, wood, leaves, bodies, and shells
Hydrosphere	Oceans, lakes, and rivers	CO_2 dissolved in water
Geosphere	Rocks and the solid Earth	Limestone rock and fossil fuels (coal, oil, and natural gas)

Carbon moves between reservoirs through **processes**:

Process	What happens
Photosynthesis	Plants and algae use sunlight to turn CO_2 into sugar, which they use to grow.
Respiration	Living things break down sugar for energy and breathe out CO_2.
Decomposition	Decomposers, like fungi (mold) and bacteria, break down dead things and release CO_2 into the air.
Combustion	Burning wood, coal, oil, or natural gas releases CO_2 into the air.
Ocean absorption	CO_2 gas from the air dissolves into ocean water. Cold water absorbs more CO_2 than warm water.
Shell building	Some ocean organisms take carbon dissolved in seawater and combine it with calcium to build hard shells and skeletons made of calcium carbonate.
Fossil fuel formation	Dead plants that are buried where there is little oxygen do not fully rot. Over millions of years, heat and pressure turn them into coal, oil, or natural gas.
Limestone formation	Shells pile up on the seafloor and get buried under layers of sediment. Over millions of years, they are pressed and cemented together into limestone rock.
Volcanic eruption	Carbon stored deep inside Earth escapes into the air as CO_2 gas when volcanoes erupt.

Fast and slow carbon: Carbon usually moves between the air, oceans, and living things in days to hundreds of years. Carbon locked in rocks and fossil fuels usually stays there for millions of years. When people burn fossil fuels, they release carbon in seconds that took millions of years to store. This is the main reason the amount of CO_2 in the atmosphere has been rising.

Part 2: Follow Count Carbonula

For each case file, read the story and use the Carbon Case File to answer.

- Circle the process that moved **Count Carbonula**.
- Fill in where **Count Carbonula** moved **from** and **to**. Choose from: Atmosphere, Biosphere, Hydrosphere, Geosphere.
- Write the code letter next to the process you chose. You will use the code letters at the end.

Case File 1: The Escape (300 million years ago)

Count Carbonula is trapped deep inside Earth in hot, melted rock. The ground shakes, a mountain explodes, and **Count Carbonula** bursts into the sky as a molecule of CO_2 gas.

Photosynthesis (W)	Volcanic eruption (S)	Decomposition (R)	Ocean absorption (A)

From: _____ To: _____ Code letter: _____

Case File 2: The Swamp

Count Carbonula drifts over a steamy swamp full of giant ferns and trees. A tree fern pulls **Count Carbonula** in through its leaves and uses sunlight to turn it into sugar, which it uses to build its trunk.

Respiration (B)	Combustion (G)	Photosynthesis (K)	Limestone formation (D)

From: _____ To: _____ Code letter: _____

Case File 3: Buried Alive

The tree fern dies and falls into the swamp mud. There is almost no oxygen in the mud, so the fern does not completely rot. Layer after layer of mud piles on top. Over millions of years, heat and pressure turn the buried plant into black, shiny coal.

Fossil fuel formation (E)	Volcanic eruption (P)	Decomposition (H)	Ocean absorption (U)

From: _____ To: _____ Code letter: _____

Case File 4: Lights On

Last year, the coal was dug out of the ground and burned at a power plant. The electricity ran the fog machines and flickering lights in a haunted house. **Count Carbonula** floated out of the smokestack as CO_2.

Photosynthesis (C)	Shell building (Y)	Ocean absorption (F)	Combustion (L)

From: _____ To: _____ Code letter: _____

Case File 5: The Rotting Jack-o'-Lantern

Count Carbonula spent the summer as part of a big orange pumpkin. After it was carved into a jack-o'-lantern, it sat on a porch for two weeks. Fuzzy mold and bacteria covered it and broke it down, and **Count Carbonula** escaped into the air as CO_2.

Volcanic eruption (M)	Decomposition (E)	Photosynthesis (J)	Fossil fuel formation (Q)

From: _____ To: _____ Code letter: _____

Case File 6: Into the Deep

On a cold, foggy night, **Count Carbonula** drifts over the North Atlantic Ocean. **Count Carbonula** dissolves into the icy seawater and sinks out of sight.

Combustion (X)	Ocean absorption (T)	Shell building (V)	Decomposition (Z)

From: _____ To: _____ Code letter: _____

Case File 7: The Bone Builder

A tiny ocean organism, smaller than a grain of sand, grabs **Count Carbonula** from the seawater. It combines **Count Carbonula** with calcium to build its hard, chalky shell.

Shell building (O)	Volcanic eruption (W)	Combustion (G)	Fossil fuel formation (B)

From: _____ To: _____ Code letter: _____

Case File 8: The Seafloor Graveyard (the future)

The tiny organism dies. Its shell sinks to the dark seafloor and lands on a pile of billions of other shells. Over millions of years, layers of sediment bury the pile and press and cement the shells into solid rock.

Respiration (A)	Combustion (P)	Ocean absorption (H)	Limestone formation (N)

From: _____ To: _____ Code letter: _____

Crack the Code

Write the code letter from each case file in its box to finish the sentence.

Count Carbonula's rock, limestone, is made from the _____s of tiny sea creatures.

1	2	3	4	5	6	7	8

Check your secret word with your teacher before you move on.

Part 3: The Planet Breathes

Scientists have measured CO_2 in the air at the Mauna Loa Observatory in Hawaii since 1958. CO_2 is measured in parts per million (ppm).

Year	1960	1970	1980	1990	2000	2010	2020	2025
CO_2 (ppm)	317	326	339	354	370	390	414	427

Source: NOAA Global Monitoring Laboratory, Mauna Loa annual averages, rounded to the nearest whole number.

1. Describe the pattern in CO_2 from 1960 to 2025.

   _____

2. How much did CO_2 increase from 1960 to 2025? Show your work.

   _____ ppm

3. CO_2 increased by 9 ppm from 1960 to 1970. How much did it increase from 2010 to 2020? Is CO_2 rising faster or slower now than in the 1960s?

   _____

4. Which case file shows the process that is the main cause of this increase? Use the Carbon Case File to explain why.

   Case File number: _____

   _____

   _____

5. **Halloween mystery:** Every year, CO_2 at Mauna Loa is lowest in late September or early October. Then it rises all through the fall and winter. Use two processes from the Carbon Case File to explain why CO_2 goes up after Halloween. (Hint: think about what happens to leaves and pumpkins in the fall.)

   _____

   _____

Part 4: Is Count Carbonula's Story Over?

Count Carbonula is now locked in limestone on the seafloor. Is this the end of **Count Carbonula**'s story? Explain why scientists call this a **cycle**. Use evidence from at least two case files in your answer.

_____

_____

Bonus Case File 9

Write a new case file for **Count Carbonula** that uses **respiration**. Your story should be 2 to 3 sentences and could include a trick-or-treater, a bat, or a candy bar. Then fill in where **Count Carbonula** moved from and to.

_____

_____

From: _____ To: _____